Topic B3: Organism level systems

B3.1 Coordination and control – the nervous system

Summary
The human nervous system is an important part of how the body communicates with itself and also receives information from its surroundings.

Underlying knowledge and understanding
Learners should have a concept of the hierarchical organism of multicellular organisms from cells to tissues to organs to systems to organisms.

Common misconceptions
Learners commonly think that their eyes see objects 'directly', like a camera, but the reality is that the image formed by the brain is based on the eyes and brains interpretation of the light that comes into the eye i.e. different people will perceive the same object or image differently. Young learners also have the misconception that some sort of 'force' comes out of the eye, enabling it to see.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM3.1i	extract and interpret data from graphs, charts and tables	M2c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
B3.1a	describe the structure of the nervous system	Central Nervous System, sensory, motor and relay neurones, sensory receptors, synapse and effectors, details of the structure of sensory and motor neurones required			Production of 3D models of neurones to illustrate their structure.
B3.1b	explain how the components of the nervous system can produce a coordinated response	it goes to all parts of the body, has many links, has different sensory receptors and is able to coordinate responses			Demonstration (by video) of someone trying to do everyday tasks whilst being given mild electric shocks (e.g. BBC Brainiac).
B3.1c	explain how the structure of a reflex arc is related to its function			M1d, WS2a, WS2b, WS2c, WS2d	Demonstration of reaction time by getting a learner to catch a falling £5 note. Research into reflexes. (PAG B6) Investigating of reaction times by ruler drop. (PAG B6)

B3.1d ☑	explain how the main structures of the eye are related to their functions	cornea, iris, pupil, lens, retina, optic nerve, ciliary body, suspensory ligaments			Demonstration of the inversion of an image through a beaker full of water. Demonstration of the features of the human eye. Investigation of eye structure by dissection. (PAG B1)
B3.1e ☑	describe common defects of the eye and explain how some of these problems may be overcome	colour blindness, short-sightedness and long-sightedness		WS2a, WS2b, WS2c, WS2d	Measurement of focal length in a variety of situations. (PAG B6) Research into eye defects, their diagnosis and treatment.
B3.1f ☑	describe the structure and function of the brain	cerebrum, cerebellum, medulla, hypothalamus, pituitary			
B3.1g ☑	**explain some of the difficulties of investigating brain function**	**the difficulty in obtaining and interpreting case studies and the consideration of ethical issues**			Discussion of problems associated with brain research including the difficulty in getting research subjects.
B3.1h ☑	**explain some of the limitations in treating damage and disease in the brain and other parts of the nervous system**	**limited ability to repair nervous tissue, irreversible damage to the surrounding tissues, difficulties with accessing parts of the nervous system**		WS1.1e, WS1.1f, WS1.1h	Research into a study of brain injury.